Paper 28.25 - N-Dimensional Game Lattice Toolkit Supplement

CQE-paper-28.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-28.25.md

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Purpose

This supplement shows how to reproduce Paper 28's local-rule game-lattice receipt. The proof is in Paper 28 and its formal verifier; full game solving is not claimed.

Digital Route

Run:

`python production/formal-papers/CQE-paper-28/verify_nd_game_lattices.py`

The expected status is `pass_with_open_obligations`. The verifier checks the forced code-tower dimensions, dimension-8 extended Hamming board, S3 move orbit, Rule 30 local emissions, forbidden carrier logging, and centroid annealing closure.

Analog Route

Draw an 8-cell board. Place a robot token at the active chart state `(1,0,1)`. Write the six S3 move cards beside the board. For each card, permute the active trace-2 state, write the target, compute the emitted bit, and draw the anneal route to a Lie-conjugate attractor.

Mark the identity carrier as forbidden with a black bar. Do not erase it; it is part of the receipt.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- admissible code-tower dimension,
- non-admissible dimension,
- legal S3 orbit move,
- forbidden carrier,
- closed anneal row,
- invalid general game-solver promotion,
- invalid real-game strategy promotion.

The answer key distinguishes finite move receipts from solved games.